The 3-adic Shadow of a Collatz Orbit:
A Friendly Guide

Companion notes to "Divergence as Sustained Compression"*

June 12, 2026

1 Where we left off

The first guide in this series ended with a picture. Playing the Collatz game (halve if even; triple-and-add-one if odd, then halve), a number can escape to infinity only by keeping the share of its "odd" steps above the critical 63.09% forever — and we know, provably, that almost every number fails to do this almost immediately, while no argument that examines numbers through any bounded lens can ever decide who succeeds. The question this paper asks is simple to state:

Suppose a number did keep it up forever. What else would have to be true about it?

The answer turns out to be startlingly concrete, and it lives in an unexpected place: not in the number's binary digits (base 2), which the first paper exhausted, but in its **base-3 digits**.

2 A magic cancellation

Run a number n through T steps of the game, of which j were odd steps, and call the result n_T. There is an exact bookkeeping identity — every multiplication and every carry accounted for, no approximations:

$$2^T \cdot n_T \;=\; 3^j \cdot n + d,$$

where the leftover term d is built up step by step along the way. Two facts about d make everything happen.

First: d **does not depend on** n. It depends only on the *pattern* of odd and even steps — the word of choices the orbit made. Two starting numbers with the same first-T pattern have exactly the same d. (Patterns and starting numbers are in one-to-one correspondence with remainders mod 2^T — that was the parity bijection of the first paper.)

Second, the magic: look at the identity *modulo* 3^j, i.e. ask only for remainders upon division by 3^j. The term $3^j \cdot n$ is divisible by 3^j — it vanishes — taking the only mention of n with it:

$$n_T \;\equiv\; (\text{something built from the pattern alone}) \pmod{3^j}.$$

*Written in collaboration with Claude (Anthropic). These notes explain the ideas; the paper and the Lean files contain the proofs. The Collatz conjecture remains open.

The shadow. After T steps, the orbit's remainder modulo 3^j — its "base-3 shadow" — is completely determined by the odd/even pattern. The starting number has no say. (Verified mechanically across twenty thousand random starts: no exceptions, as the theorem demands.)

3 Minting coins with a small ledger

Now count, because the two sides of this equation are counted in different currencies.

How many shadows could there be? The shadow lives among the remainders modulo 3^j, and there are 3^j of those. For a surviving orbit, j is locked to about 63.09% of T by the critical-line results — and here sits the coincidence that powers the whole paper. Measured in bits, 3^j amounts to $j \cdot \log_2 3 \approx 0.6309 \times 1.585 \times T$ bits, and

$$0.6309\ldots \times 1.585\ldots = 1 \quad \text{exactly.}$$

A surviving orbit mints **exactly one bit of base-3 information per step**. Not approximately — the two famous constants of this problem multiply to precisely 1.

How many patterns could be doing the minting? Patterns with 63% odd steps are rare: coin-flip counting (Pascal's triangle again) says there are only about $2^{0.95\,T}$ of them — 0.95 bits per step of pattern.

So the mint produces 1.00 bits per step of base-3 currency, printed from a ledger containing only 0.95 bits per step of instructions. The currency must be *redundant*: out of the 3^j possible shadows, surviving orbits can only ever land on an exponentially tiny fraction — about $2^{-0.045\,T}$ of them. At a window of just 18 steps this is already visible by brute force: surviving orbits hit only 2.17% of the available remainders modulo 3^{12}.

The reduction. A number that escapes to infinity must emit base-3 digits that are about 4.5% compressible — carrying provably less information than their length — at every scale, by consistent amounts, forever. "Divergence" and "a number that out-compresses its own arithmetic" are the same thing.

Both halves of this are now machine-verified theorems: the density floor (escapees are pinned to the critical 63.09%, where the mint runs at exactly 1 bit per step) and the confinement count (the shadow set is small, with — pleasingly — the *same* constants 17/27 and 3^T that ran the "almost all numbers fall" theorem in the first paper; the base-2 and base-3 mirrors are one phenomenon).

4 The ledger has a grammar: scales talk to each other

The leftover term d obeys a beautiful composition rule. If you run a window of length S and then a window of length T, with patterns w_1 and w_2,

$$d(w_1 w_2) \;=\; 3^{j_2}\, d(w_1) \;+\; 2^S\, d(w_2)\,:$$

the second window's base-3 weight scales the first ledger entry, the first window's base-2 length scales the second. (Readers who know some dynamics will recognize a skew product of the doubling and tripling worlds; everyone else can read it as: *the books at different scales are not independent — they're nested.*)

The experiments confirm this with force. At a window of 18 steps, knowing an orbit's full-window shadow pins down its half-window shadow almost uniquely (about 1.09 candidates on average, out of hundreds possible). A diverging number wouldn't just need a compressible shadow at each scale separately — it would need all its shadows, at all scales, to *cohere* through this grammar. That consistency requirement is a constraint nobody has ever exploited, and it is the paper's first recommendation for future attack.

One more experimental finding sharpens the hunt. The slowest possible escapees — those hugging the critical line — are forced into extremely regular patterns (so-called Sturmian words, the most orderly aperiodic sequences there are). Feeding such patterns through the shadow map, they cluster dramatically: 400 of them produced only 167 distinct shadows, where 400 random patterns of the same density produced 400. The most dangerous escape routes live in the *thinnest* part of the base-3 world — precisely where arithmetic has the most leverage to forbid them.

5 So is it solved?

No — and the paper is explicit about what stands in the way. The remaining statement has a familiar shape: it asks, in essence, whether a whole number's base-2 behavior and base-3 behavior can conspire — whether an integer can have such special joint structure in the two bases that it sustains the compression. Mathematics has a famous circle of problems about exactly this ("$\times 2$, $\times 3$" problems, after Furstenberg), and they are notoriously hard: even the innocent-sounding claim *"the binary digits of* 3^k *look random"* is open. Two of our own earlier results explain why no shortcut exists: every finite behavior really occurs (so the compressed shadow sets are never empty — counting alone cannot finish), and no bounded-memory argument can decide who inhabits them.

What the paper contributes is the conversion. Before: "prove no number escapes," a problem with no agreed-upon shape. After: a single quantitative statement — *no integer compresses its own base-3 digit stream by* 4.5% *at every scale consistently forever* — with machine-checked constants, a numeric gap between 0.95 and 1.00 that does not depend on any estimate being sharp, three identified levers (cross-scale consistency, structured-pattern exclusion, and the ergodic question for the rigidity specialists), and an honest map of which lever sits in known-hard territory.

6 Summary in four sentences

An exact identity makes an orbit's remainder modulo 3^j a function of its odd/even pattern alone — the starting number cancels out. Escaping numbers are pinned to the critical density, where they mint exactly one bit of base-3 information per step from patterns worth only 0.95 bits per step — so their base-3 digits must be compressible by about 4.5%, at every scale, forever, and the shadows at different scales are locked together by an exact composition law. All of this is machine-verified; the experiments show the true compression is even stronger, the scales nearly determine each other, and the most regular escape patterns cluster in the thinnest shadow sets. The Collatz conjecture remains open — but its divergence half now has a precise shape: it asks whether an integer can beat the entropy of its own arithmetic.